SIMATS ENGINEERING

CO3- AT2 - Design Task - Medium

COURSE NAME: Advance Wireless Communication Systems /

5G / 6G Communication Systems

COURSE CODE: ECA56

COURSE OUTCOME COVERED

CO: 3 - *Design spectrum access strategies, waveform generation methods, and multiple access techniques for efficient wireless transmission systems.* (BL6)

Assessment Tool Weightage: 35%

Short description about assessment: Students design a system / model based on given requirements to assess creativity and application skills.

Dr.

QUESTIONS: 1 (1 X 100 = 100) **Total Marks: 100 Duration: 3hrs**

1.Design a satellite-assisted communication model using Low Earth Orbit satellite for remote wireless coverage.

Code of Conduct:

I ____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Design Task					
Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Understanding of Design	20%	Demonstrates deep understanding of design objectives, constraints, and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Lacks understanding of the design context
Evaluation of Design	25%	Provides in-depth critique identifying strengths, weaknesses, and design gaps	Identifies key issues with reasonable analysis	Limited critique; mostly descriptive feedback	No meaningful critique provided
Justification	20%	Supports critique with strong reasoning, evidence, and relevant principles	Provides justification with some supporting evidence	Weak or unclear justification	No justification for feedback
Suggestions for Improvement	20%	Offers innovative, practical, and well-justified improvement suggestions	Provides useful suggestions with minor gaps	Suggestions are limited or partially relevant	No constructive suggestions provided
Communication	15%	Communicates feedback clearly, professionally, and engages actively in discussion	Communicates well with minor issues	Communication lacks clarity or engagement	Poor communication and minimal participation

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clc;
clear;
close all;

%%
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% SATELLITE-ASSISTED COMMUNICATION USING LEO SATELLITE
% Remote Wireless Coverage Model
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%% 1. SYSTEM PARAMETERS

c = 3e8;                    % Speed of light (m/s)
fc = 2e9;                   % Carrier frequency = 2 GHz
lambda = c/fc;              % Wavelength

Pt = 30;                    % Transmit power (dBm)
Gt = 15;                    % Transmit antenna gain (dBi)
Gr = 20;                    % Receive antenna gain (dBi)

Bandwidth = 20e6;           % Channel bandwidth = 20 MHz
NF = 5;                     % Receiver noise figure (dB)

LEO_altitude = 600e3;       % LEO satellite altitude = 600 km
Earth_radius = 6371e3;      % Earth radius = 6371 km

%% 2. NOISE POWER

k = 1.38e-23;               % Boltzmann constant
T = 290;                    % Temperature in Kelvin

Noise_W = k*T*Bandwidth;
Noise_dBm = 10*log10(Noise_W*1000) + NF;

fprintf('============================================\n');
fprintf(' LEO SATELLITE ASSISTED COMMUNICATION MODEL\n');
fprintf('============================================\n\n');

fprintf('Carrier Frequency : %.2f GHz\n',fc/1e9);
fprintf('Bandwidth         : %.2f MHz\n',Bandwidth/1e6);
fprintf('LEO Altitude      : %.0f km\n',LEO_altitude/1000);
fprintf('Noise Power       : %.2f dBm\n\n',Noise_dBm);
```

```
%% 3. REMOTE USER DISTANCE FROM SATELLITE

% Ground coverage radius
coverageRadius = 1000e3;    % 1000 km

% Generate user locations
theta = linspace(0,2*pi,100);

x = coverageRadius*cos(theta);
y = coverageRadius*sin(theta);

%% 4. SLANT RANGE CALCULATION

% Approximate slant distance between satellite and users
% Satellite is directly above center of coverage area

d = sqrt(LEO_altitude^2 + x.^2 + y.^2);

%% 5. FREE SPACE PATH LOSS

FSPL = 20*log10(4*pi*d/lambda);

%% 6. RECEIVED POWER

Pr = Pt + Gt + Gr - FSPL;

%% 7. SNR

SNR_dB = Pr - Noise_dBm;

%% 8. DATA RATE USING SHANNON CAPACITY

SNR_linear = 10.^(SNR_dB/10);

Capacity = Bandwidth .* log2(1 + SNR_linear);

Capacity_Mbps = Capacity/1e6;
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```
%% 9. DISPLAY RESULTS

fprintf('Coverage Radius = %.0f km\n',coverageRadius/1000);

fprintf('Maximum Path Loss = %.2f dB\n',max(FSPL));

fprintf('Minimum Received Power = %.2f dBm\n',min(Pr));

fprintf('Minimum SNR = %.2f dB\n',min(SNR_dB));

fprintf('Minimum Data Rate = %.2f Mbps\n',min(Capacity_Mbps));

fprintf('Maximum Data Rate = %.2f Mbps\n\n',max(Capacity_Mbps));

%%
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% 10. COVERAGE MAP
%
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figure;

plot(x/1000,y/1000,'LineWidth',2);
hold on;

plot(0,0,'p','MarkerSize',15,'LineWidth',2);

grid on;
axis equal;

xlabel('Distance East-West (km)');
ylabel('Distance North-South (km)');

title('LEO Satellite Remote Wireless Coverage');
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legend('Coverage Boundary','Satellite Projection');

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% 11. PATH LOSS VS DISTANCE
%
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distance_km = sqrt(x.^2 + y.^2)/1000;

figure;

plot(distance_km,FSPL,'LineWidth',2);

grid on;

xlabel('Ground Distance from Satellite Projection (km)');
ylabel('Free Space Path Loss (dB)');

title('LEO Satellite Path Loss');

%%
================================================================
=====
% 12. RECEIVED POWER VS DISTANCE
%
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figure;

plot(distance_km,Pr,'LineWidth',2);

grid on;
```

```
xlabel('Ground Distance (km)');
ylabel('Received Power (dBm)');

title('Received Power in LEO Satellite Communication');

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% 13. SNR VS DISTANCE
%
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figure;

plot(distance_km,SNR_dB,'LineWidth',2);

grid on;

xlabel('Ground Distance (km)');
ylabel('SNR (dB)');

title('SNR vs Ground Distance');

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% 14. DATA RATE VS DISTANCE
%
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figure;

plot(distance_km,Capacity_Mbps,'LineWidth',2);
```

```
grid on;

xlabel('Ground Distance (km)');
ylabel('Data Rate (Mbps)');

title('Achievable Data Rate vs Ground Distance');

%%
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% 15. 3D COVERAGE VISUALIZATION
%
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figure;

[X,Y] = meshgrid(-1000:50:1000,-1000:50:1000);

D = sqrt(LEO_altitude^2 + ...
        (X*1000).^2 + ...
        (Y*1000).^2);

FSPL_3D = 20*log10(4*pi*D/lambda);

Pr_3D = Pt + Gt + Gr - FSPL_3D;

surf(X,Y,Pr_3D);

xlabel('East-West Distance (km)');
ylabel('North-South Distance (km)');
zlabel('Received Power (dBm)');

title('3D LEO Satellite Coverage Performance');

colorbar;
shading interp;
```

```
grid on;

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% 16. DIFFERENT LEO ALTITUDES
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altitudes = [400 500 600 700 800 1000];  % km

centerDistance = altitudes*1000;

FSPL_altitude = 20*log10( ...
   4*pi*centerDistance/lambda);

Pr_altitude = Pt + Gt + Gr - FSPL_altitude;

SNR_altitude = Pr_altitude - Noise_dBm;

SNR_linear_altitude = 10.^(SNR_altitude/10);

Capacity_altitude = ...
   Bandwidth .* log2(1 + SNR_linear_altitude);

%% Plot altitude comparison

figure;

plot(altitudes,Capacity_altitude/1e6,'-o','LineWidth',2);

grid on;

xlabel('LEO Satellite Altitude (km)');
ylabel('Data Rate (Mbps)');
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title('Effect of LEO Satellite Altitude on Data Rate');

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% 17. PRINT ALTITUDE RESULTS
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fprintf('LEO ALTITUDE ANALYSIS\n');
fprintf('---------------------------------------------\n');
fprintf('Altitude(km)   PathLoss(dB)   SNR(dB)   Rate(Mbps)\n');

for i = 1:length(altitudes)

   fprintf('%8.0f      %8.2f      %6.2f     %8.2f\n', ...
      altitudes(i), ...
      FSPL_altitude(i), ...
      SNR_altitude(i), ...
      Capacity_altitude(i)/1e6);

end

fprintf('\n============================================\n');
fprintf('Simulation Completed Successfully!\n');
fprintf('============================================\n');
```
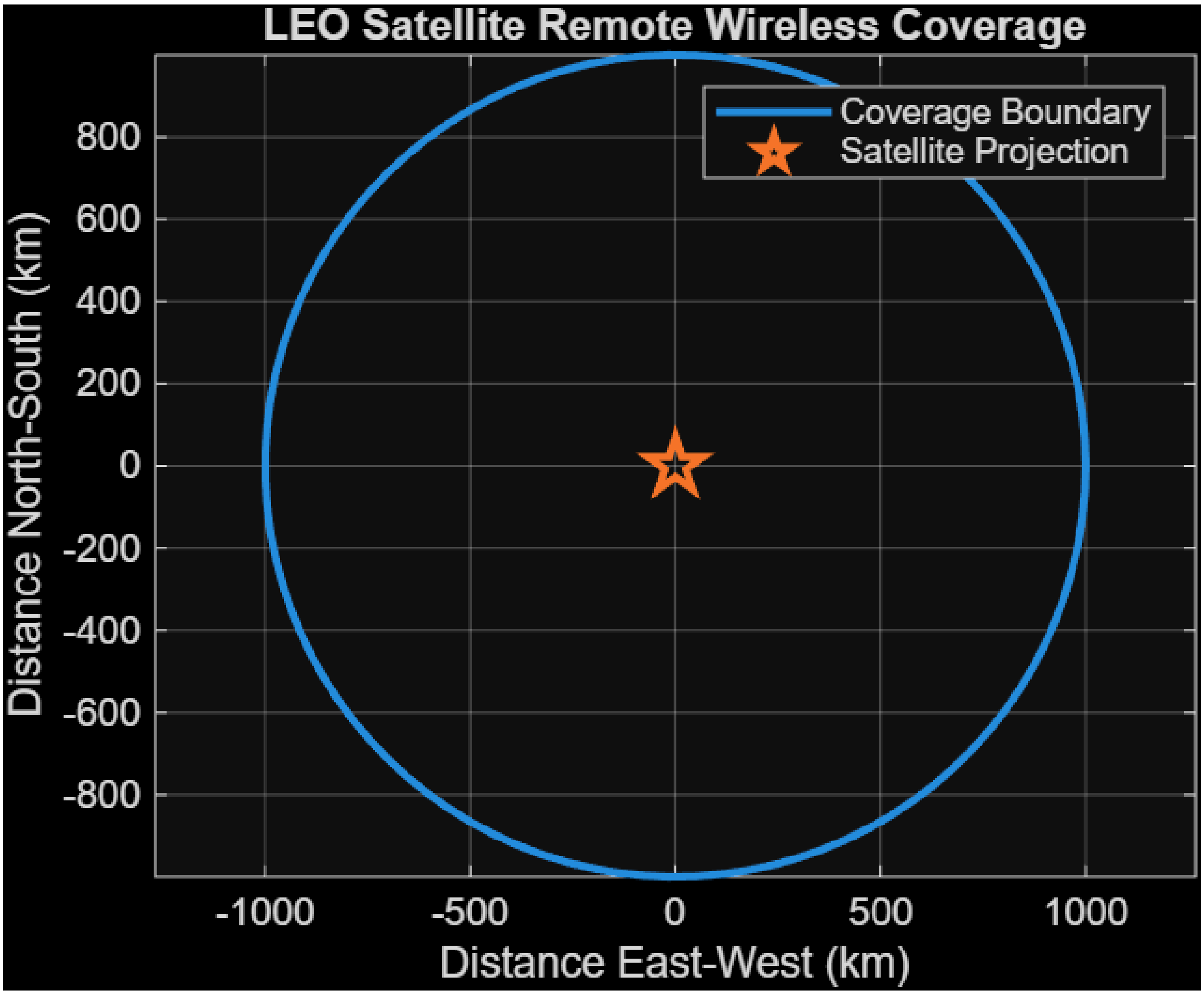
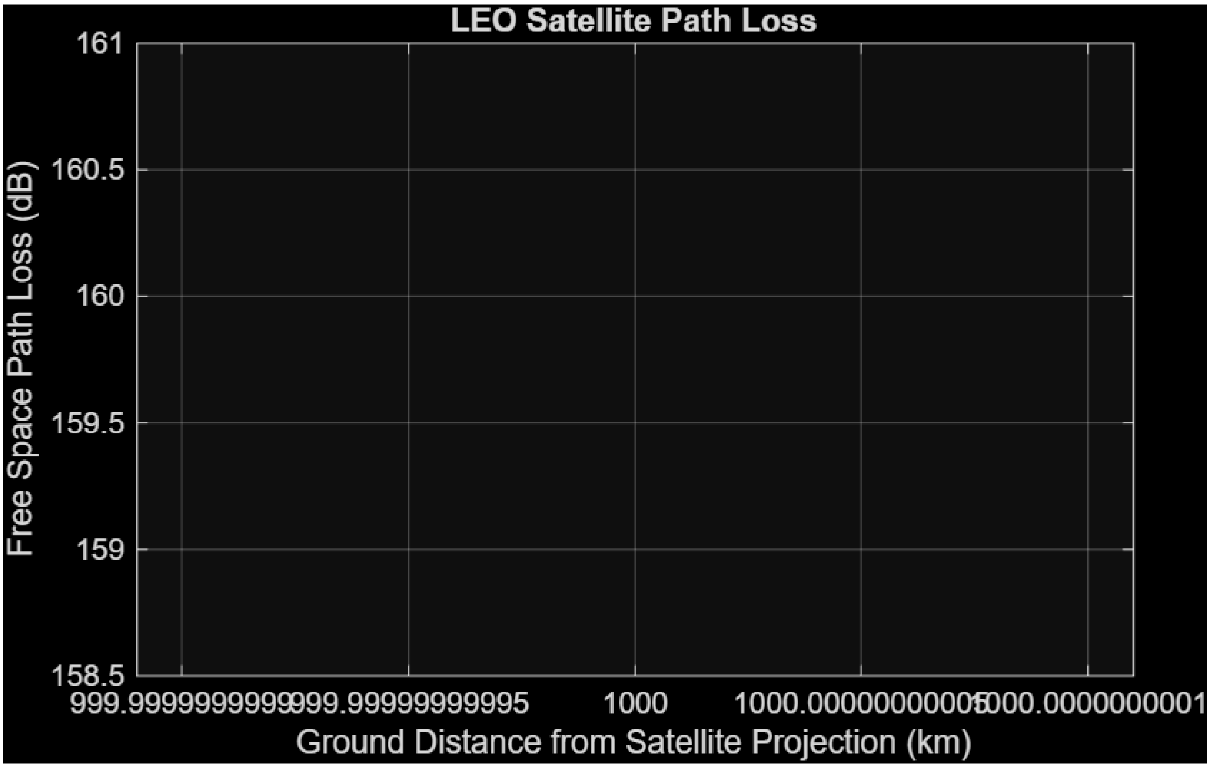
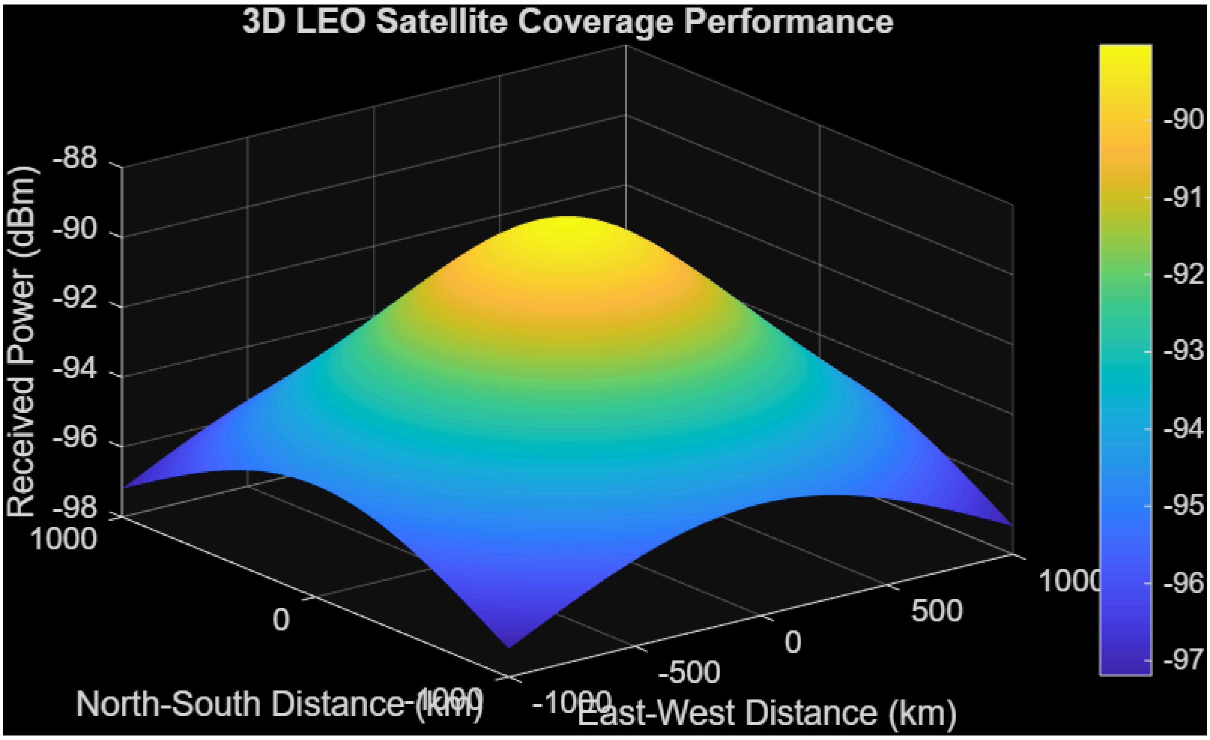
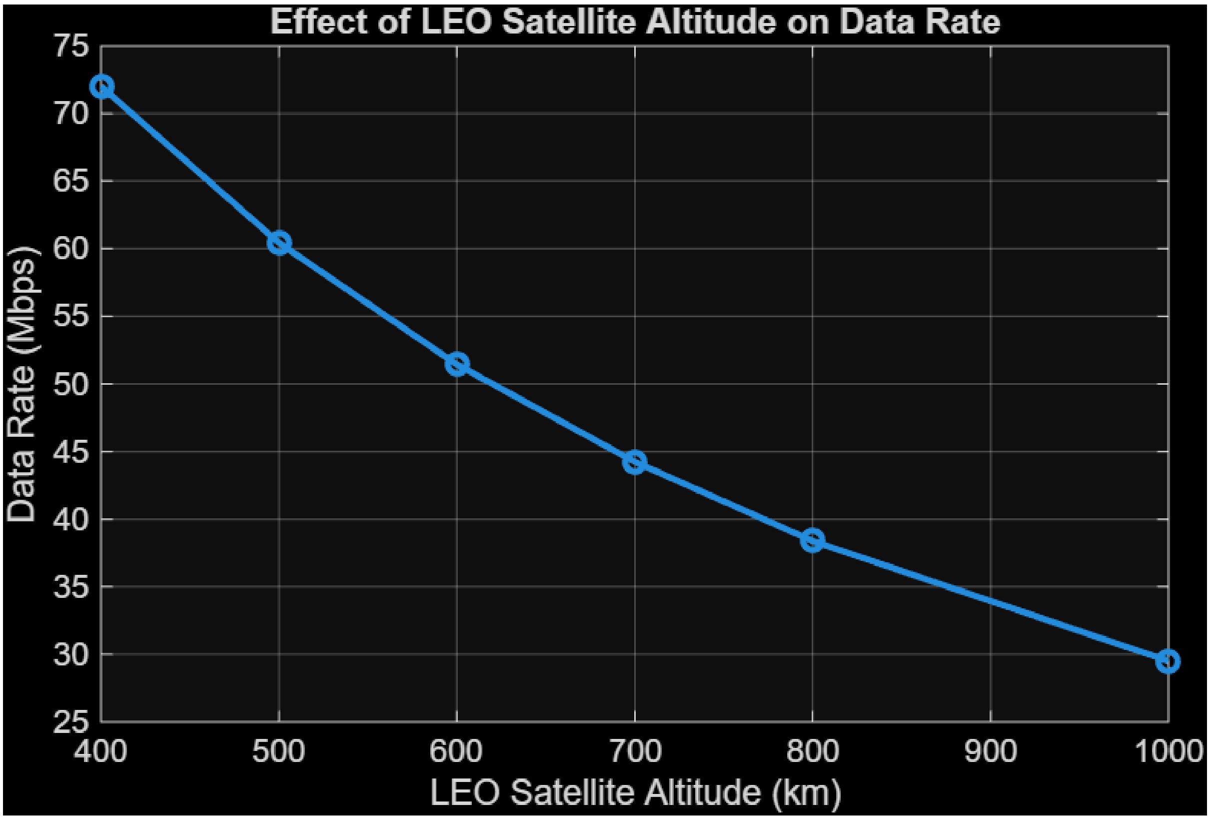


LEO Satellite Remote Wireless Coverage
Coverage Boundary
Satellite Projection
800
600
400
200
0
-200
-400
-600
-800
Distance North-South (km)
-1000
-500
0
500
1000
Distance East-West (km)


LEO Satellite Path Loss
161
160.5
160
159.5
159
158.5
Free Space Path Loss (dB)
999.9999999999
999.9999999995
1000
1000.0000000005
1000.0000000001
Ground Distance from Satellite Projection (km)


3D LEO Satellite Coverage Performance
Received Power (dBm)
-88
-90
-92
-94
-96
-98
1000
0
-1000
North-South Distance (km)
-1000
-500
0
500
1000
East-West Distance (km)
-90
-91
-92
-93
-94
-95
-96
-97


Effect of LEO Satellite Altitude on Data Rate
Data Rate (Mbps)
75
70
65
60
55
50
45
40
35
30
25
400
500
600
700
800
900
1000
LEO Satellite Altitude (km)